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RESEARCH ARTICLE

Ensuring Zero Trust Security in Consumer Internet of Things Using Federated Learning-Based Attack Detection Model

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ABSTRACT The rising adoption of Consumer Internet of Things (CIoT) devices brings heightened concerns about security vulnerabilities and privacy breaches. Given their connection to daily routine and data collection abilities, it is vital to protect individual data against unauthorized entry and possible leaks proactively. Data protection's maximum communication and storage charge will gradually challenge the typical ecology of integrated over-the-cloud learning and handling for IoT environments. Federated learning (FL) is developed as the most reasonable method for this challenge. In FL, training data-driven machine learning (ML) approaches is a form of association among several users with no need for data carried to a fundamental point; therefore, improving communication and storage charge offers a great degree of consumer-level secrecy. FL is an advanced ML that performs appropriately by inherently prioritizing confidentiality, circumventing requirements for centralized data collection, and supporting the defence. FL improves security in IoT by enabling decentralized model training without centralizing sensitive data, preserving privacy, and reducing communication costs. Integrating FL into attack detection models ensures zero-trust security, minimizing vulnerabilities and enhancing protection against cyber threats while keeping data on-device. This study develops an enhanced artificial gorilla troop optimizer with an FL-based attack detection and classification (EGTO-FLADC) approach on the CIoT environment. The EGTO-FLADC approach aims to improve security in the CIoT environment via FL and the attack detection process. Primarily, the EGTO-FLADC technique utilizes the FL model for collaborative training of combined models with many devices with no data-sharing process. In addition, the EGTO-FLADC technique uses a min-max scalar for data normalization. For attack detection, the EGTO-FLADC technique employs the temporal convolution networks with a gated recurrent unit (TCN-GRU) model and its hyperparameters are chosen by the EGTO model. The EGTO technique involves the integration of the conventional GTO model with the marine predator's algorithm (MPA) model. The performance evaluation of the EGTO-FLADC approach is examined under the EdgeIIoTset dataset. The experimental validation of the EGTO-FLADC approach portrayed a superior accuracy value of 97.11% over existing models.

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• **INDEX TERMS** Consumer Internet of Things, federated learning, attack detection, marine predator algorithm, DDoS attacks.

I. INTRODUCTION

The fast proliferation of uncertain user IoT devices and simple where the attackers can find them utilizing online services like Shodan delivers an ever-growing pool of attack sources [1]. By involving arrays of these exposed IoT devices, invaders can execute huge-scale attacks like Distributed Denial of Service (DDoS), spamming, and phishing beside sources on the Internet [2]. The growth in IoT-based DDoS attacks perceived currently will be possible to endure until IoT companies take charge and produce gadgets with safety naturally constructed. The IoT has the potential to become a novel method for upcoming cyber-attacks and thus provides numerous tasks [3]. An enlarging amount of DDoS attacks search for influence user-level IoT devices, the problems emphasized earlier, united by an absence of expert knowledge or recognition of inherent exposures by holders of these devices, offers a challenge and region of examination for safety investigators [4]. Botnets are an assortment of devices inspired by general malware in which a command-and-control structure permits a bot controlling to launch malicious attacks. Once infected, bots will begin scanning the Internet to discover equally vulnerable devices [5]. Hackers mainly target IoT devices in the consumer market owing to their plentiful exposure and the absence of awareness of device safety [6]. In 2016, the most tarnished IoT-specific botnet, recognized as Mirai, emerged. Containing over 400,000 negotiated IoT devices, the Mirai botnet introduced significant DDoS attacks, attaining a rate of transmission of 1.1Tbps. Targeting networked user-based IoT devices, the explosive progress and offensive abilities of the Mirai botnet promoted the accurate breeding ground for equally shattering botnets, with novel worries of malware being identified daily [7].

The extensive spread and growth of massive botnet movements underlines the need to investigate Internet-scale mitigation and remediation methods. In addition, consumer devices are at a high risk of being used by malware owing to the absence of information technology and safety departments supervising the safety postures of such gadgets [8]. As it is viewed, no skill or specific department object is answerable for observing, recovering, and averting IoT device usage in user realms. Consumer information is in danger of being developed without appropriate events to safeguard IoT devices, obviously over-inspecting the malicious objects that are vigorously emerging and spreading malware to take advantage of poorly protected devices [9]. Recently, FL has developed as a dispersed privacy-upholding resolution to finding this persistent requirement. In FL, training in ML methods for data-driven uses is a collaborative performance among dispersed customers without merging the consumer information. FL's collaborative and distributed nature is a standard fit for the system edge [10]. Furthermore,

the raw data gathered at every IoT device is not spread to anyone; FL delivers an effective device to safeguard consumer confidentiality, mainly in the IoT area. IoT sensors might arrest information regarding consumers who have privacy-sensitive data. FL ensures privacy by keeping sensitive consumer data on IoT devices, enabling secure model training without sharing raw data.

This study develops an enhanced artificial gorilla troop optimizer with an FL-based attack detection and classification (EGTO-FLADC) approach on the CIIoT environment. The EGTO-FLADC approach aims to improve security in the CIIoT environment via FL and the attack detection process. Primarily, the EGTO-FLADC technique utilizes the FL model for collaborative training of combined models with many devices with no data-sharing process. In addition, the EGTO-FLADC technique uses a min-max scalar for data normalization. For attack detection, the EGTO-FLADC technique employs the temporal convolution networks with a gated recurrent unit (TCN-GRU) model and its hyperparameters are chosen by the EGTO model. The EGTO technique involves the integration of the conventional GTO model with the marine predator's algorithm (MPA) model. The performance evaluation of the EGTO-FLADC approach is examined under the EdgeIIoTset dataset. The key contribution of the EGTO-FLADC approach is listed below.

- The EGTO-FLADC technique utilizes FL for collaborative model training across several devices. It ensures data privacy by eliminating the requirement for data sharing and improves overall performance by enabling effective and secure training while preserving user confidentiality.
- The EGTO-FLADC model employs min-max scalar-based normalization to standardize the data, ensuring consistency and improving the training process. This methodology improves the method's capability to learn effectively from diverse data sources while maintaining performance accuracy.
- The approach significantly enhances performance in sequence prediction tasks by integrating TCN and GRU models. This incorporation implements the merits of both models to capture long-range dependencies and improve predictive accuracy.
- The hyperparameters of the TCN-GRU model are optimized using the EGTO technique, which effectively combines GWO with the MPA. This hybrid approach enhances optimization accuracy and improves the model's overall performance.
- The novelty of the EGTO-FLADC method is in the hybridization of EGTO with MPA, improving the optimization process for the TCN-GRU model's hyperparameters. This hybrid model improves attack detection accuracy and efficiency in IoT by using

the merits of both methods to tackle dynamic IoT challenges.

The article is structured as follows: Section II presents the literature review, Section III outlines the proposed method, Section IV details the results evaluation, and Section V concludes the study.

II. RELATED WORKS

In [11], a differential privacy (DP) with FL (DPFL) structure with adaptive clipping, which utilizes a Gaussian mechanism, is developed. At the same time, conventional FL is based on a central server and is vulnerable to solitary opinions of malicious and failure node attacks. The clear broadcast of transitional parameters can lead to the inference of private data. So, a FL method dependent upon a blockchain (BC) is developed to improve safety, decentralization, and fairness. In [12], a FL-based intrusion detection system (IDS) called FELIDS is presented. Notably, the FELIDS model defends data privacy over local learning. In contrast, the gadgets take advantage of the data of their peers by distributing only upgrades from their method with an accumulation server that delivers an enhanced recognition technique. The FELIDS model uses 3-DL classifiers such as deep neural network (DNN), convolutional neural network (CNN), and recurrent neural network (RNN). In [13], a FL model is used that is both popular and cost-effective. The consumers in the FL technique use their confidentially kept data and procedures for forecasting and calculating method variations after getting a primary ML method from the server. This effort recommends an FL-based model to execute matrix factorization over cooperative filtering. Lin et al. [14] suggest an advanced federated DL model named Fed-PEMC. This technique unites local difference confidentiality and model compression methods. By using reinforcement learning (RL) models, namely deep RL (DRL) for compression, Fed-PEMC decreases the dimension of the technique. The technique also presents a modified label sample to quicken the model training. Before uploading the technique, local variance privacy protection is executed using the compacted method.

Tiwari et al. [15] present the multi-agent federated RL policy (MFRLP) specified in fog and mobile agents dependent upon the socket remote procedure call (RPC) model. The main aim is to project a user-centric cyborg-effective training method that performs complete use of devices with the least stays in the IoMT method. The research improves the RPC dependent upon RL and FL, which implements dynamic variations in the atmosphere for cyborg use. In [16], a consistent privacy-preserving FL-based DL service structure is presented. FL diminishes the discretion problems of the traditional cooperative learning method by combining numerous locally skilled methods. This structure presents an edge and cloud-powered service-oriented structure recognizing the main modules for remaining systems-based Federated DL (FDL) with different discretion for making reliable locally skilled methods. The service method spoils the capability of the complete FDL

procedure as services guarantee reliable implementation over privacy protection. Lastly, the technique progresses to a privacy preservative local method accumulation device for FDL. Xu et al. [17] projected an energy-effectual classified FL model and expressed a multiple-objective optimizer issue with diverse resources and device association. Since network state data is inaccessible for every round using the Lyapunov optimizer structure, the importance of local methods is reformulated. In addition, a low-complexity dual-operation gadget association model is presented. Gupta et al. [18] projected an advanced structure for safe and privacy-preserving decentralized FL tailored to modified recommendations within the user electronic segment. To achieve this, the technique harnesses the homomorphic encryption power. Moreover, the model influences the BC technique to find a protected, dispersed data management and exchange basis.

Mudassir et al. [19] propose high-performing DL methods for classifying botnet attacks targeting IIoT devices and networks, aiming to improve cybersecurity in Industry 4.0 environments. Al-Garadi et al. [20] provide a comprehensive survey of ML and DL methods for enhancing the security of IoT systems. Alwahedi et al. [21] provide a comprehensive overview of ML applications in cyber threat detection for IoT environments, analyze advanced ML-based IDSs, and explore future advancements with Generative AI and large language models to enhance IoT security. Javeed et al. [22] present a horizontal FL model integrating CNN and Bidirectional Long Short-Term Memory (BiLSTM) for improved intrusion detection (ID) in IoT. CNN extracts spatial features, while BiLSTM captures temporal dependencies. The model follows a zero-trust approach, sharing only learned weights with a centralized FL server. Rashid et al. [23] propose an FL method for ID in IoT networks, ensuring privacy by training local data on devices and sharing only parameter updates with a central server. Rabieinejad et al. [24] propose a two-level privacy-preserving framework integrating FL with partially homomorphic encryption to improve security and protect user privacy in CIoT devices. Popoola et al. [25] develop FDL models for ID in CIoT using relevant datasets. Hussain et al. [26] present a secure data-sharing framework integrating local DP (LDP), FL, and a permissioned BC in a zero-trust environment enhanced by IPFS and cryptographic hash functions. Rani et al. [27] explore the potential of FL for attack detection in the Internet of Vehicles (IoV) and detect the key challenges for its real-world deployment in smart cities and transportation systems. Zhu et al. [28] propose a secure framework integrating FL, BC, and zero trust principles for CIoT, using secret sharing for privacy-preserving aggregation. Table 1 summarizes the studies on zero trust security in consumer IoT with FL for ID.

The existing FL models for IoT security, while concentrating on privacy preservation, face challenges in scalability, specifically in large-scale IoT systems. These models often depend on unrealistic network data assumptions that may not reflect the dynamic nature of real-world IoT environments. In addition, they tend to overlook various attack types

TABLE 1. Summary of studies on zero trust security in consumer IoT with FL for ID.

Reference Number	Objective	Method	Dataset	Measures
[11]	To propose an AI-driven DPFL framework	DPFL with Adaptive Clipping Gaussian Mechanism BC-based FL	Benchmark dataset	Standard measures
[12]	To propose a FELIDS for securing agricultural-IoT infrastructures while ensuring data privacy.	FL for Data Privacy Local Learning with Model Updates DNN CNN RNN	CSE-CIC-IDS2018 MQTTset InSDN	Benchmark measures
[13]	To propose a FL-based recommender system for privacy-preserving, efficient, and accurate personalized recommendations in consumer electronics.	FL Matrix Factorization through Collaborative Filtering User-Prediction Model Updates Data Privacy Preservation via Federated Approach	Benchmark Recommender System Data	RMSE Accuracy Computational and Communication Overhead
[14]	To propose Fed-PEMC for improved privacy, accuracy, and efficiency in consumer electronics using DP and model compression.	FDL Local DP Model Compression via DRL Customized Label Sampling Privacy Protection via DP (ϵ, δ)	MNIST CIFAR-10	Testing Accuracy DP Communication Efficiency Model Size Reduction Performance Improvement
[15]	To design a cyborg-efficient training system for IoMT using FRL to reduce delays.	MFRLP RPC paradigm RL and FL integration	Benchmark dataset	Benchmark measures
[16]	To propose a trustworthy privacy-preserving FL-based DL service framework for Industrial IoT systems.	FL DL DP Edge and Cloud-powered Service-oriented Architecture Residual Networks-based FDL Privacy-preserving Local Model Aggregation	Standard dataset	Performance evaluation through experiments Privacy preservation in model aggregation
[17]	To propose an energy-efficient hierarchical FL model to reduce latency and optimize training performance in consumer electronics.	Energy-efficient hierarchical FL Multi-objective optimization problem formulation Lyapunov optimization framework Low-complexity two-operation device association algorithm Resource allocation	Benchmark dataset	Training latency reduction Energy consumption control Learning performance comparison with benchmarks Numerical results showing performance improvement
[18]	To propose a secure, privacy-preserving decentralized FL framework for personalized recommendations in consumer electronics.	Homomorphic encryption for data privacy BC for decentralized data management and integrity FL for client collaboration	Real-world consumer electronics data	Privacy preservation System transparency Malicious activity prevention Recommendation accuracy
[19]	To enhance cybersecurity, propose DL models for classifying botnet attacks in Industrial IoTs (IIoT) systems.	ANN LSTM GRU	IIoT malware attack datasets (specific datasets not mentioned)	Accuracy, Precision, Recall, F1-Score

TABLE 1. (Continued.) Summary of studies on zero trust security in consumer IoT with FL for ID.

[20]	To provide a comprehensive survey of ML and DL methods for enhancing security in IoT systems.	ML DL	IoT security threat datasets (specific datasets not mentioned)	IoT system attack detection Security enhancement evaluation
[21]	To provide an overview of ML methods for IoT cyber threat detection, trends, challenges, and future advancements.	ML-based IDS Comparative analysis of state-of-the-art ML-based IDSs	IoT security datasets (specific datasets not mentioned)	Cyber threat detection accuracy Comparison of ML-based IDS performance
[22]	To propose a horizontal FL-based IDS combining CNN and BiLSTM to enhance IoT security.	FL CNN BiLSTM	CICIDS2017 Edge-IIoTset	Accuracy Precision Recall AUC Loss
[23]	To propose a FL-based method for ID in IoT networks to ensure privacy and security.	FL Local IoT device data training Parameter updates are shared with the global server	Edge-IIoTset	Accuracy Precision Recall F1-Score
[24]	To propose a privacy-preserving framework using FL and partially homomorphic encryption to secure CIoT systems.	FL Partially Homomorphic Encryption Privacy-preserving model aggregation	Benchmark dataset	Attack prediction error rates False alarm reduction
[25]	To develop FDL models for privacy-preserving CIoT ID with improved performance and efficiency.	FL DL Binary and Multi-class Classification	Three recent and relevant datasets with diverse attack types	Accuracy Precision Recall F1-Score Training Time Reduction
[26]	To propose a secure data-sharing framework using LDP, FL, and BC to protect sensitive IoT data.	LDP FL Permissioned BC IPFS Cryptographic Hash Functions	Standard dataset	Latency Throughput Privacy Accuracy Transaction Efficiency
[27]	To propose an FL approach for high-accuracy attack detection in the IoV networks.	FL Attack Detection in IoV Networks	Benchmark dataset	Accuracy Precision Recall F1-Score
[28]	To propose a FL framework with BC and zero-trust for secure CIoT collaboration.	FL BC Integration (Main-Side BC) Zero Trust Principles Privacy-Preserving Aggregation (Secret Sharing) Resampling-Based Robust Aggregation	Standard dataset	Model Accuracy Convergence Speed Resilience to Byzantine Attacks

and classification scenarios, which limits their ability to handle diverse and evolving threats. Privacy-preserving mechanisms, such as homomorphic encryption and DP, can degrade system performance and model accuracy, mitigating their practical efficiency in resource-constrained environments. A significant research gap is in enhancing the real-time, adaptive capabilities of FL models to detect and reduce new attack patterns as they increase. Current models

face difficulty in addressing the complexities of dynamic attack scenarios, specifically in IoT and IoV environments. Furthermore, while incorporating FL with BC is an exciting approach, issues like high network latency, system efficiency, and the requirement for optimized communication protocols remain underexplored. The lack of research on balancing privacy, computational efficiency, and model performance in decentralized networks also limits the broader application

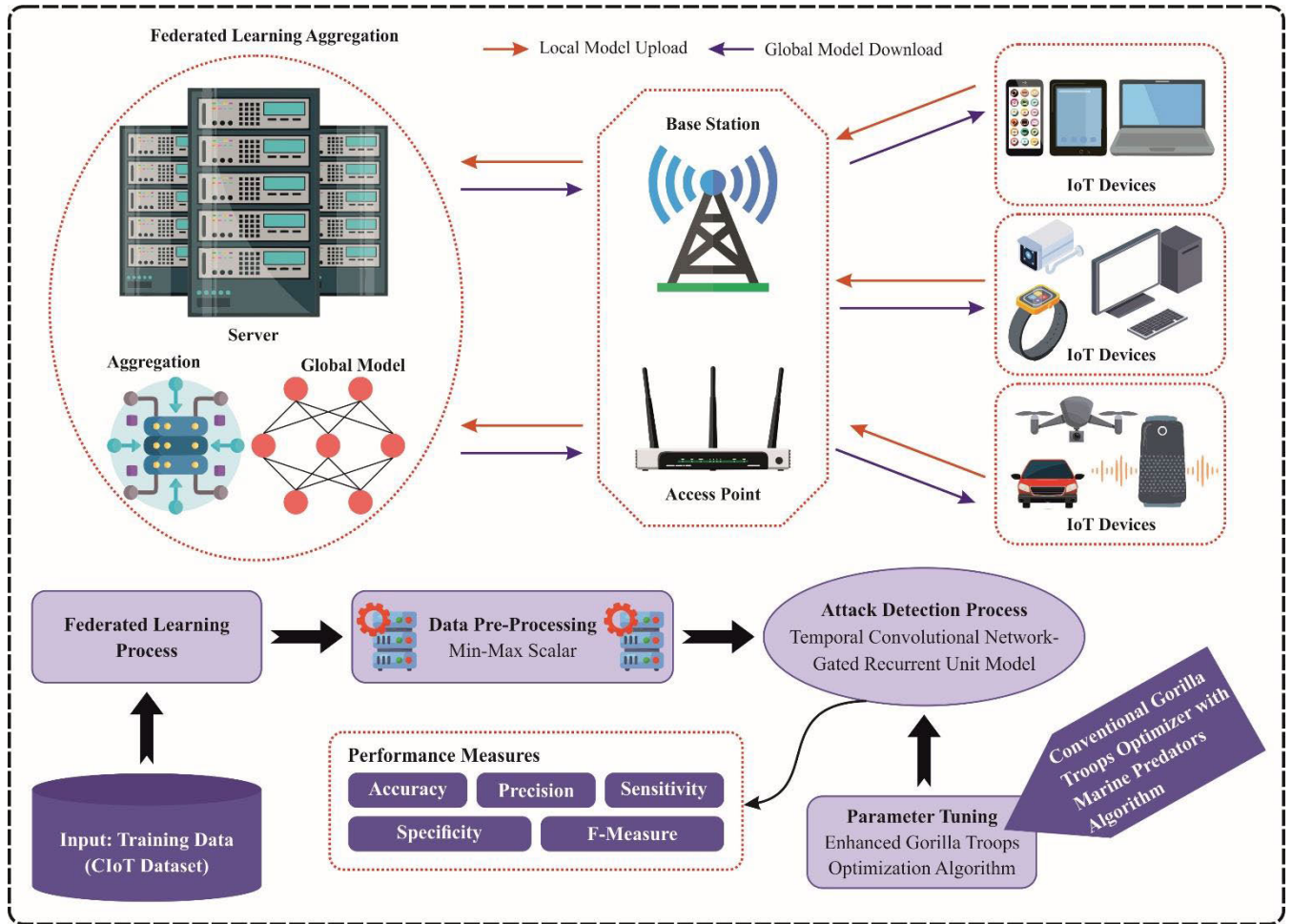


FIGURE 1. The overall flow of the EGTO-FLADC model.

of FL in security solutions. More studies are needed to evaluate FL models in real-world deployment scenarios and address the practical challenges of maintaining decentralized, privacy-preserving systems across heterogeneous IoT devices. Researchers should concentrate on developing standardized frameworks for securely aggregating data, enhancing model robustness, and improving the overall performance of FL-based systems in dynamic, real-time IoT ecosystems.

III. THE PROPOSED METHODOLOGY

This study proposes an EGTO-FLADC approach to a CIoT environment. The EGTO-FLADC approach aims to boost security in the CIoT environment via FL and the attack detection process. It contains four major FL procedures for ID: min-max-based normalization, TCN-GRU-based attack recognition, and EGTO-based hyperparameter fine-tuning. Fig. 1 signifies the workflow of the presented EGTO-FLADC model. Algorithm 1 illustrates the EGTO-FLADC model.

A. UTILIZING FL FOR ID PROCESS

Primarily, the EGTO-FLADC technique applies the FL concept for collaborative training of combined models with many devices with no data-sharing process. Depending on the distribution of data among dissimilar parties, FL is categorized into feature-based (vertical) FL, FTL, and horizontal (sample-based) FL [29]. The data share parallel sample space but different feature space during the vertical of feature-based FL. The dataset varies in the model, but feature space has a lesser intersection during FTL. The database shares a parallel feature space but differs in data samples during the horizontal or sample-based FL. In the proposed model, a horizontal FL is considered. The two standard designs of FL are the top one (centralized) and the bottom one (decentralized). At first, every partner in the data-sharing association gathers local datasets from the data acquisition resource (i.e., vehicle sensors or surveillance cameras) for centralized design. Then, the raw data was utilized for the local anomaly recognition system, which is well-trained and framed with available data. Next, the local method was synchronized occasionally with the cloud server. At the same

Algorithm 1 EGTO-FLADC Model**Input:**

- devices_data: Data from various devices for FL
- training_data: Training dataset for model training
- training_labels: Labels corresponding to the training data
- test_data: Testing dataset
- test_labels: Labels corresponding to the test data

Output:

- Model accuracy on the test dataset

Step 1: Initialize Federated Learning

1. Initialize the FL model.
2. For every device in devices_data:
 - Train the technique on local data.
3. Combine the models from all devices to create a global model.
4. Return to the global model.

Step 2: Data Normalization using Min-Max Scaling

1. Initialize Min-Max Scaler.
2. Apply Min-Max normalization to training_data to scale the data to a fixed range [0, 1].
3. Return normalized training data.

Step 3: Build Hybrid TCN-GRU Model

1. Initialize the TCN model.
2. Initialize GRU model.
3. Integrate both models to create a hybrid model.
4. Return the hybrid TCN-GRU model.

Step 4: Hyperparameter Optimization using EGTO and MPA

1. Initialize EGTO with MPA for optimization.
2. Use the EGTO-MPA optimizer to find optimal hyperparameters for the hybrid model.
3. Update the model with the optimized hyperparameters.
4. Return the optimized model.

Step 5: Model Training

1. Train the hybrid model utilizing the normalized training_data and training_labels.
2. Return the trained model.

Step 6: Model Evaluation

1. Use the trained model to predict the labels of test_data.
2. Compare the predicted labels with test_labels to calculate accuracy.
3. Return the model accuracy.

Step 7: Execute the Process

1. Load devices_data, training_data, training_labels, test_data, and test_labels.
2. Perform **Step 1**: Federated Learning to get the global model.
3. Perform **Step 2**: Normalize the training_data.
4. Perform **Step 3**: Build the hybrid TCN-GRU model.
5. Perform **Step 4**: Optimize model hyperparameters utilizing EGTO-MPA.
6. Perform **Step 5**: Train the model.
7. Perform **Step 6**: Evaluate the model and print the accuracy.

End of Algorithm

time, the bottom of FL does not rely on the central cloud server but permits a direct exchange model and synchronizes parameters among the partners. On the other hand, these two techniques have potential, and the desired approach might depend on the partner involved in the data-sharing federation.

FL is intended for the collaborative training of aggregated methods with many nodes or devices without sharing raw data [30]. Its advantages over centralized DL and ML models are as follows: •Keep raw information in the owner: In FL,

clients or devices may retain personal information by training local models without sharing user data with the central server.

- Mitigate data scarcity: FL may access more diverse and much larger datasets and assist in overcoming the problem of limited data on some devices. This will improve the client's ability to predict unknown items and the generalization capability of the aggregated technique.
- Train with heterogeneous data: FL allows a training model with a considerable amount of data (sample

quantity, distribution, and features, along with the data type that might vary) from various sources. FL demonstrates its efficacy in model development in achieving efficient anomaly detection, realizing privacy-enhancing FL-based ID, and overcoming unknown attacks. Furthermore, FL has promoted several more IDSs, emphasizing its potential as an efficacious cybersecurity and IoT security strategy.

B. NORMALIZATION USING MIN-MAX SCALAR

Next, the EGTO-FLADC technique uses a min-max scalar for data normalization [31]. This effective data normalization technique transforms features into a fixed range, typically [0, 1], ensuring uniformity in the input data for ML models. Its primary advantage is that it preserves the relationships between the original data values while eliminating potential biases caused by varying scales of features. This is crucial in models like TCN-GRU, which optimize performance and depend on consistent input ranges. Min-max scaling is preferred over other techniques, such as Z-score normalization, as it doesn't assume a Gaussian distribution, making it more versatile for diverse datasets. Moreover, it ensures that features with different magnitudes or units contribute equally to the model, improving the training process and enhancing convergence speed.

Normalization is also a data preprocessing technique to scale numerical features within a fixed range, typically from zero to one. It operates by deducting the lowest value and dividing it by the range (the highest value minus the lowest value). This maintains a similar connection between the data points while safeguarding that the feature is on an identical scale, which is more advantageous for the algorithm sensitive to feature scaling, like a gradient descent-based approach. Min-max scaling is widely applied in ML tasks where the magnitude of features could differ, ensuring that features equally contribute to the analysis without being exceedingly influenced by its scale.

C. ATTACK DETECTION USING TCN-GRU METHOD

In this stage, the EGTO-FLADC model uses the TCN-GRU technique for attack detection [32]. This model is chosen because it captures long-term dependencies and local patterns in time-series data. TCN outperforms handling sequential data using causal convolutions, effectively allowing the model to capture temporal dependencies without traditional RNNs' limitations. Meanwhile, the GRU improves the capability of the method to manage vanishing gradients, making it appropriate for learning from long data sequences. Together, the TCN-GRU model integrates the merits of both architectures, giving enhanced accuracy and effectiveness in detecting intrinsic attack patterns in real time. This hybrid methodology outperforms traditional methods like simple feedforward networks or basic RNNs, which may need help with long-range dependencies or real-time performance. Fig. 2 illustrates the architecture of the TCN-GRU model.

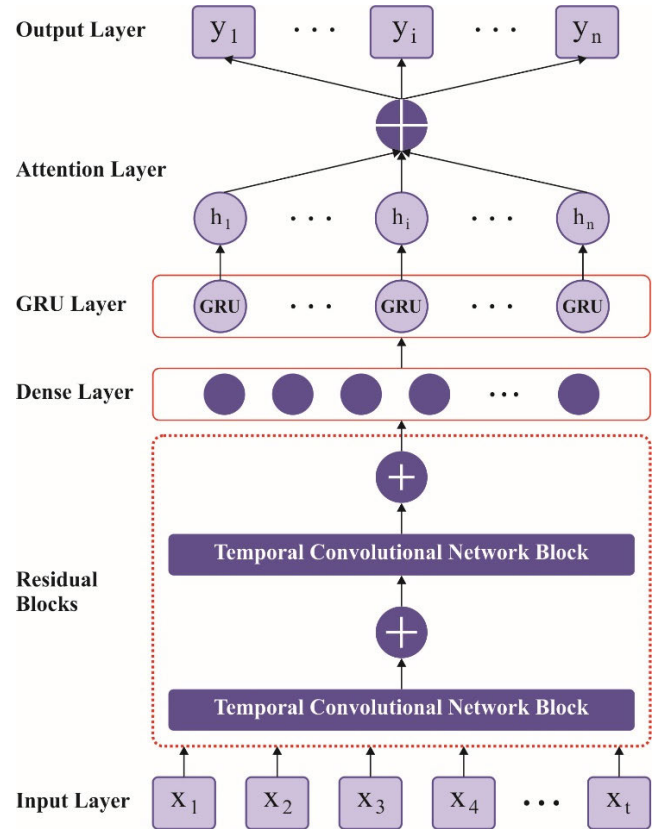


FIGURE 2. The architecture of the TCN-GRU model.

TCN is a special type of CNN for time series data processing. TCN integrates two dissimilar NN structures of GRU. The key feature of typical convolution with dilation convolution ensures that data from the present time and the predecessor are employed while forecasting the output at present. In contrast, the dilation convolutional layer can cover long input sequences between convolutional kernels to take long-term dependency. As a kind of RNN, GRU performs remarkably well in capturing short-term dependency in time sequence data. This integrates the TCN's long-term learning ability with the time-based dynamics exhibiting the ability of GRUs.

In TCN-GRU architecture, TCN initially procedures an input series to take the long-term dependency, and the GRU layer receives the TCN output to process the short-term dependency in the time-sequence data.

$$F(t) = (X \cdot dW)(t) \quad (1)$$

In Eq. (1), the dilated convolutional function is represented as $\ast_d$, the input sequence is formulated by X , the convolution kernel is indicated as W , and the time step is indicated by t .

The reset and update gates of GRU are given as follows:

Update the door:

$$z_t = \sigma(W_z x_t + U_z h_{t-1}) \quad (2)$$

Reset the door:

$$r_t = \sigma(W_r x_t + U_r h_{t-1}) \quad (3)$$

New status:

$$\tilde{h} = \tanh(W x_t + U(r_t \odot h_{t-1})) \quad (4)$$

Output status:

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h} \quad (5)$$

where h_t refers to the hidden layer at the existing time, the sigmoid activation function is represented as σ , $Error::0x0000$ indicates the entry-wise multiplication, and the weight matrices are considered W and U .

GRU effectively captures unexpected events, predicts short-term transformations, and handles dynamic changes and short-term dependencies in time-series data. In contrast, the TCN-GRU improves feature learning by addressing limitations when capturing long-range dependencies. The fusion model also mitigates the gradient vanishing problem and demonstrates more stable gradient propagation related to classical RNNs.

D. HYPERPARAMETER TUNING PROCESS

Finally, the EGTO method is utilized for hyperparameter selection, which involves integrating the conventional GTO model with MPA. A novel meta-heuristic model stimulated by gorilla troops' social intellect naturally is known as Artificial GTO [33]. This model accurately expresses gorillas' group life, and novel devices are intended to execute exploitation and exploration.

1) EXPLORATION STAGE

Throughout this stage, three different operators were used to improve the search: one tangled affecting an unfamiliar position, pointing to research deeper into the GTO model. The second part is the changeover to other gorillas, improving exploitation and exploration stability. In the meantime, the 3rd part relates to the exploration stage; travelling to a usual location suggestively increases the GTO model's capability to discover various developing spaces. These different operators are representatively signified by the below-mentioned calculation:

$$GX(t+1) = \begin{cases} (ub - lb) \times r_1 + lb, & \text{if } rand < z \\ (r_2 - C) \times X_r(t) + D \times B, & \text{if } rand \geq 0.5 \\ X(i) - D \times (D \times (X(t) - GX_r(t)) \\ \quad + r_3 \times (X(t) - GX_r(t))) & \text{if } rand < 0.5 \end{cases}$$

where $C = (\cos(2 \times r_4) + 1) \times \left(1 - \frac{it}{Maxit}\right)$

$$\begin{aligned} D &= C \times k \\ B &= E \times X(t) \\ E &= [-C, C] \end{aligned} \quad (6)$$

Here, lb and ub represent the lower and upper limits correspondingly. $GX(t+1)$ means the gorilla candidate's

location in the subsequent iteration. z refers to a parameter that ranges from 0 to 1. $X(t)$ signifies the present vector of the gorilla location. $r_1, rand, r_2, r_3$, and r_4 denote the randomly generated values ranging from zero to one. k represents a randomly produced value that ranges from -1 to 1 . $X_r(t)$ indicates a follower of the gorillas arbitrarily nominated from the complete gorillas and $GX_r(t)$.

2) EXPLOITATION STAGE

Throughout this stage, dual behaviours or approaches are used. The 1st approach, recognized as Follow the Silverback, was used if a certain state $C \geq W$ has happened. Where W signifies a pre-defined parameter fixed previously in the optimizer procedure. Accurately, the 1st tactic is calculated as below:

$$\begin{aligned} GX(t+1) &= D \times M \times (X(t) - X_{silverback}) \\ &\quad + X(t) \text{ where } M \\ &= \left(\left| \frac{1}{N} \sum_{i=1}^N GX_i(t) \right|^g \right)^{\frac{1}{g}}, \\ g &= 2^D \end{aligned} \quad (7)$$

$GX(t+1)$ signifies the upgraded gorilla location in the searching space at the subsequent iteration, N indicates the complete no. of gorillas, and $X_{silverback}$ denotes the finest solution. By including this approach in the optimizer procedure, the model's primary goal is to exploit capable areas of the searching space by manipulating data from the finest solution that has been met up to now. This tactic improves the model's ability to meet near-optimum solutions throughout exploitation. Generally, Eq. (7) plays a vital part in directing the exploitation and exploration, manipulating the model's performance in discovering optimum solutions.

The 2nd device is the Competition for grownup females and employed if $C < W$. This device was computed as given below:

$$\begin{aligned} GX(i) &= X_{silverback} - (X_{silverback} \times Q - X(t) \times Q) \\ &\quad \times A \text{ where } Q = 2 \times r_5 - 1, \text{ and } A = \beta \\ H &= \begin{cases} N_1, & rand \geq 0.5 \\ N_2, & rand < 0.5 \end{cases} \end{aligned} \quad (8)$$

wherein β denotes a parameter to be assumed value before the optimizer process. r_5 represents the randomly produced value that ranges from zero to one.

The enhanced GTO is named higher and lower velocity percentages dependent upon the MPA model. This method has been intended to resolve the prospect of the optimal value, which may fall into local goals. Numerous current models have been proposed dependent upon this tactic, and they are used for multiple optimizer issues with the parameter of the organizer unite proportional-integral-derivative (PID) and slight order control models utilizing Enhanced Runge Kutta Optimizer and optimum dimension and position of numerous FACTS mechanisms to attain diminishing cost of fuel and

reducing power losses. This development was based on dual phases. The 1st phase is the ratio of higher velocity. The accurate method is given below:

$$it < \frac{1}{3}Maxit \quad (9)$$

$$S = \vec{R}_B \otimes (E - \vec{R}_B \otimes X_i(t)) \quad (10)$$

$$X_i(t+1) = X_i(t) + P \cdot CF \otimes S \quad (11)$$

Here, $\vec{R}_B$ means a vector of randomly generated numbers from the normal distribution, which imitates Brownian motion. $\otimes$ mentions to entry-wise multiplication. The novel position was pretended by multiplying R_B the preceding location, $P=0.5$ refers to the constant value, and R_B embodies a vector of even randomly generated values in the interval of [0 and1]. This state occurs throughout the 1st and 2nd of iterations if the size of the step is huge, typically a higher level of search capability. The fittest solution (E) was chosen as the finest position to create a matrix by the mentioned expression:

$$E = \begin{bmatrix} Xb'_{1,1} & \dots & Xb'_{1,d} \\ \vdots & \ddots & \vdots \\ Xb'_{n,1} & \dots & Xb'_{n,d} \end{bmatrix} \quad (12)$$

Xb is the finest solution that is copied n times to form an E matrix. n refers to the number of search agents, and d signifies dimensions. The 2nd phase is the ratio of lower velocity. This phase occurs near the finish of the optimizer process and is classically linked to higher exploitation ability. Lévy is the finest method for lower-velocity ratio. It was signified below:

$$it > \frac{1}{3}Maxit \quad (13)$$

$$S = \vec{R}_L \otimes (\vec{R}_L \otimes E - X_i(t)) \quad (14)$$

$$X_i(t+1) = E + P \cdot CF \otimes S \quad (15)$$

The EGTO model invents a fitness function (FF) to achieve the enhanced solution of the classifier. It describes an optimistic amount to signify the improved performance of the candidate solution. Here, minimizing the classifier error rate has been measured as FF as defined in Eq. (16).

$$\begin{aligned} fitness(x_i) &= Classifier\ Error\ Rate(x_i) \\ &= \frac{No.of\ misclassified\ instances}{Total\ no.of\ instances} \times 100 \end{aligned} \quad (16)$$

IV. EXPERIMENTAL VALIDATION

The performance analysis of the EGTO-FLADC technique is studied under the EdgeIoTset dataset [34]. The dataset includes normal and attack samples, as defined in Table 2. A total of 2500 normal samples are involved. In addition, the attack types include DDoS-UDP (C2), DDoS-ICMP (C3), SQL-injection (C4), DDoS-TCP (C5), Vulnerability scanner (C6), Password (C7), DDoS-HTTP Uploading (C8), Backdoor (C9), Port-scanning (C10), XSS (C11), Ransomware (C12), Fingerprinting (C13), and MITM (C14).

The suggested technique is simulated using the Python 3.6.5 tool on a PC with an i5-8600k, 250GB SSD, GeForce 1050Ti 4GB, 16GB RAM, and 1TB HDD. The parameter settings are: learning rate: 0.01, activation: ReLU, epoch count: 50, dropout: 0.5, and batch size: 5.

TABLE 2. Details on dataset.

Type of Event	Event Labels	No. of Record
Normal	C1	2500
DDoS-UDP	C2	2500
DDoS-ICMP	C3	2500
SQL-injection	C4	2500
DDoS-TCP	C5	2500
Vulnerability scanner	C6	2500
Password	C7	2500
DDoS-HTTP Uploading	C8	2500
Backdoor	C9	2500
Port-scanning	C10	2500
XSS	C11	2500
Ransomware	C12	2500
Fingerprinting	C13	2500
MITM	C14	2500
Total No. of Record		35000

The performance of DL classification models is examined by using diverse evaluation metrics. Eq. (17) represents $accu_y$, measuring the overall proportion of correct predictions, while Eq. (18) computes $prec_n$, the ratio of correct positive predictions. Eq. (19) defines $sens_y$, evaluating the capability of the model to detect true positives correctly, and Eq. (20) represents $spec_y$, which measures the model's capacity to identify true negatives. Finally, Eq. (21) shows the $F_{measure}$, integrating precision and sensitivity into a single metric to balance their trade-offs, especially when dealing with imbalanced classes. These metrics provide a comprehensive understanding of model performance with the following equations:

$$Accu_y = \frac{TP + TN}{TP + TN + FP + FN} \quad (17)$$

$$Prec_n = \frac{TP}{TP + FP} \quad (18)$$

$$Sens_y = \frac{TP}{TP + FN} \quad (19)$$

$$Spec_y = \frac{TN}{TN + FP} \quad (20)$$

$$F_{Measure} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \quad (21)$$

Here, TP represents True Positives, TN specifies True Negatives, FP portrays False Positives, and FN denotes False Negatives. These metrics allow for a detailed assessment of the model's accuracy and efficiency in handling various classification tasks, including class imbalances.

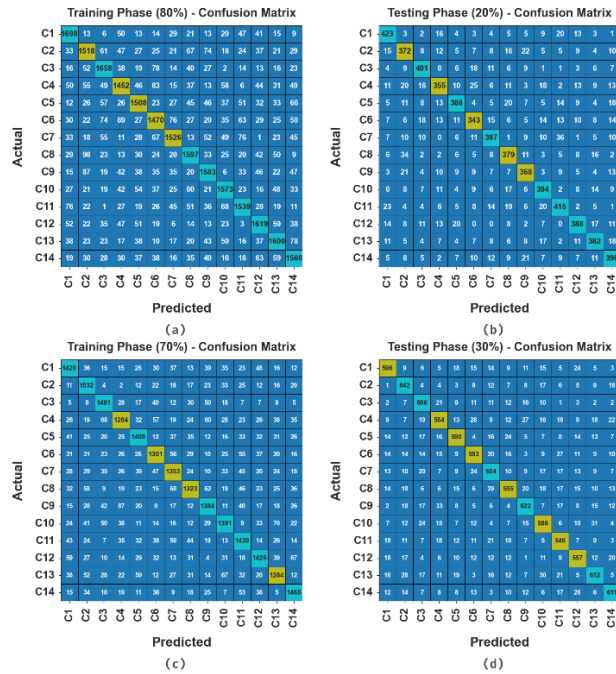


FIGURE 3. Confusion matrices of (a-c) 80% and 70% of TRPH and (b-d) 20% and 30% of TSPH.

Fig. 3 establishes the confusion matrices formed by the EGTO-FLADC method under 80:20 and 70:30 of TRPH/TSPH. The results state that the EGTO-FLADC approach has effectual classification and detection of all 14 classes accurately.

The attack recognition outcomes of the EGTO-FLADC model are investigated with 80:20 of TRPH/TSPH in Table 3 and Fig. 4. The outcome values signify the proficient efficiency of the EGTO-FLADC model. With 80%, the EGTO-FLADC approach displays an average $accu_y$ of 96.89%, $prec_n$ of 78.22%, $sens_y$ of 78.22%, $spec_y$ of 98.32%, and $F_{measure}$ of 78.19%. Additionally, with 20%, the EGTO-FLADC approach exhibits an average $accu_y$ of 96.72%, $prec_n$ of 77.05%, $sens_y$ of 77.04%, $spec_y$ of 98.23%, and $F_{measure}$ of 77.01%.

The performance of the EGTO-FLADC approach is graphically projected in Fig. 5 in the form of validation $accu_y$ (VAL) and training $accu_y$ (TRA) curves below 80:20 of TRPH/TSPH. The figure displays a valuable explanation of the behaviour of the EGTO-FLADC approach over numerous epochs, signifying its learning process and generalized abilities. Remarkably, the figure completes a steady growth in the TRA and VAL with epoch growth. It confirms the adaptive nature of the EGTO-FLADC technique in the pattern detection procedure on both TRA and TES data. The rising trend in VAL condenses the capability of the EGTO-FLADC technique to familiarize the TRA. Also, it shines in delivering exact identification of unseen data, directing out the strong generalized abilities.

Fig. 6 discloses a complete representation of the TRA loss (TRALOS) and VAL loss (VALLOS) results of the

TABLE 3. Attack recognition outcome of EGTO-FLADC model under 80:20 of TRPH/TSPH.

Classes	$Accu_y$	$Prec_n$	$Sens_y$	$Spec_y$	$F_{measure}$
TRPH (80%)					
C1	97.42	79.79	85.37	98.35	82.49
C2	96.53	75.64	75.82	98.12	75.73
C3	97.14	78.65	82.49	98.27	80.52
C4	96.45	75.98	73.04	98.24	74.48
C5	96.76	77.93	75.82	98.36	76.86
C6	96.34	75.42	72.88	98.16	74.13
C7	97.08	81.52	76.41	98.67	78.88
C8	96.91	78.02	79.33	98.27	78.67
C9	96.87	78.60	78.06	98.34	78.33
C10	97.02	79.44	78.69	98.43	79.07
C11	97.06	79.62	78.20	98.49	78.90
C12	97.10	79.05	80.91	98.35	79.97
C13	97.03	79.17	79.60	98.38	79.38
C14	96.73	76.21	78.43	98.13	77.30
Average	96.89	78.22	78.22	98.32	78.19
TSPH (20%)					
C1	97.07	78.33	82.78	98.20	80.49
C2	96.10	71.68	74.70	97.74	73.16
C3	97.31	80.20	81.84	98.48	81.01
C4	96.14	75.85	69.34	98.26	72.45
C5	96.86	80.00	75.93	98.51	77.91
C6	96.34	74.73	71.01	98.22	72.82
C7	96.86	78.82	76.94	98.40	77.87
C8	96.59	74.31	77.82	97.99	76.03
C9	96.89	76.35	77.97	98.25	77.15
C10	96.77	76.80	78.64	98.17	77.71
C11	96.64	77.86	78.01	98.18	77.93
C12	97.00	79.67	77.76	98.48	78.70
C13	96.94	78.28	77.96	98.37	78.12
C14	96.57	75.81	77.89	98.04	76.83
Average	96.72	77.05	77.04	98.23	77.01

EGTO-FLADC method over dissimilar epochs under 80:20 of TRPH/TSPH. The advanced decrease in TRALOS highlights the EGTO-FLADC method, enhancing the weights and reducing the TRA and TES data identification error. The figure specifies the EGTO-FLADC model's suggestion with the TRA data, emphasizing its ability to seize patterns within both databases. The EGTO-FLADC model repeatedly enhances its parameters to decrease the differences between the forecast and real TRA classes.

Examining the precision-recall (PR) curve, as exposed in Fig. 7, the results confirmed that the EGTO-FLADC method gradually attains enhanced PR values over all classes under 80:20 of TRPH/TSPH. It confirms the boosted abilities of the

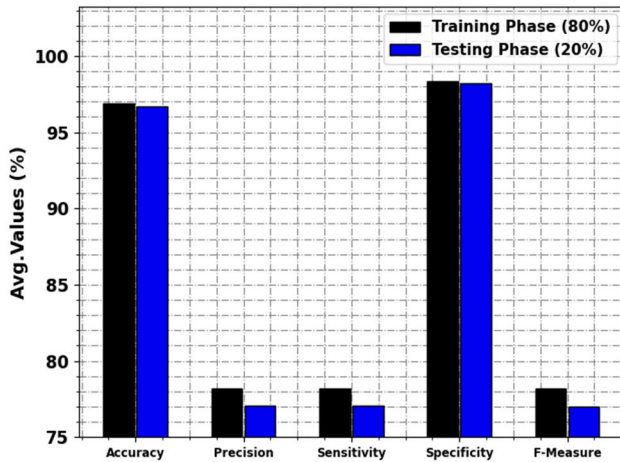


FIGURE 4. The average outcome of the EGTO-FLADC method under 80:20 of TRPH/TSPH.

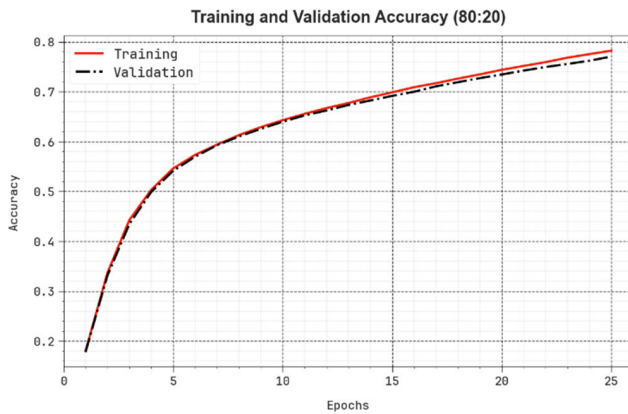


FIGURE 5. acc_y curve of EGTO-FLADC technique under 80:20 of TRPH/TSPH.

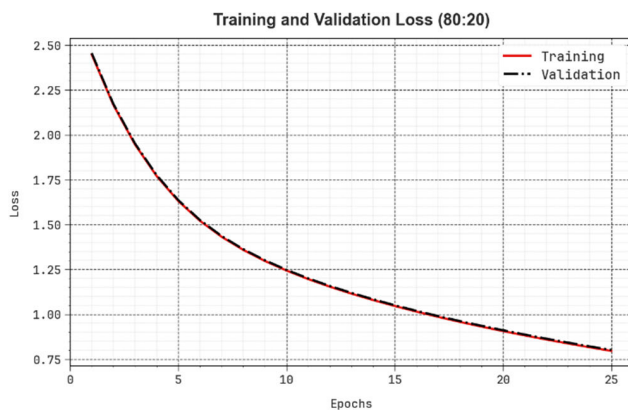


FIGURE 6. Loss curve of EGTO-FLADC technique below 80:20 of TRPH/TSPH.

EGTO-FLADC technique in classifying different classes and displaying the ability of the detection class labels.

Likewise, in Fig. 8, ROC curves moulded by the EGTO-FLADC technique outdid the classification of different labels

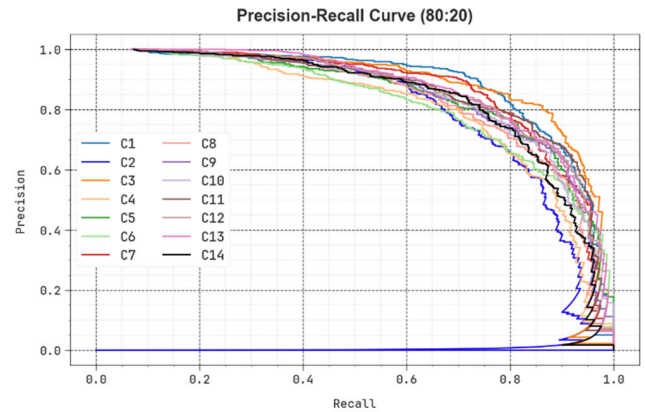


FIGURE 7. PR curve of EGTO-FLADC model under 80:20 of TRPH/TSPH.

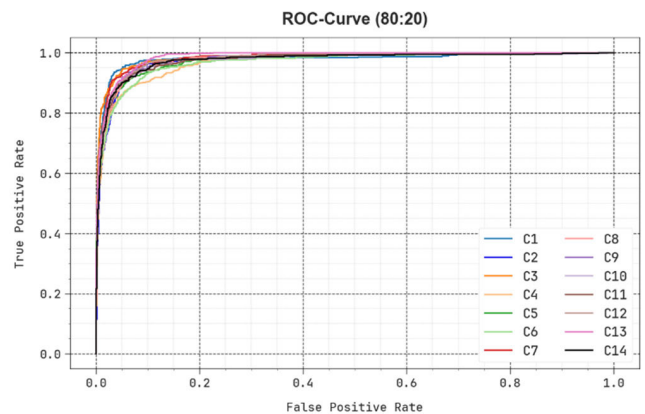


FIGURE 8. ROC curve of EGTO-FLADC model under 80:20 of TRPH/TSPH.

under 80:20 of TRPH/TSPH. This thoroughly explains the trade-off between TPR and FRP over distinct threshold and epoch count detection values. The figure highlights the enhanced classifier outcomes of the EGTO-FLADC method under each class, outlining its efficacy in addressing numerous classification issues.

The attack recognition results of the EGTO-FLADC technique are examined with 70:30 of TRPH/TSPH in Table 4 and Fig. 9. The attained values indicate the proficient efficacy of the EGTO-FLADC technique. With 70%, the EGTO-FLADC technique shows an average acc_y of 97.11%, $prec_n$ of 79.79%, $sens_y$ of 79.79%, $spec_y$ of 98.45%, and $F_{measure}$ of 79.76%. Furthermore, with 30%, the EGTO-FLADC model exhibits an average acc_y of 97.05%, $prec_n$ of 79.36%, $sens_y$ of 79.38%, $spec_y$ of 98.41%, and $F_{measure}$ of 79.33%.

The performance of the EGTO-FLADC method is graphically presented in Fig. 10 in the custom of TRA acc_y and VAL acc_y curves under 70:30 of TRPH/TSPH. The outcome displays a suitable understanding of the behaviour of the EGTO-FLADC method over numerous epochs, indicating its learning procedure and generalized abilities. Notably, the figure gathers a constant development in the TRA acc_y and VAL acc_y with growth in epochs. It confirms the adaptive

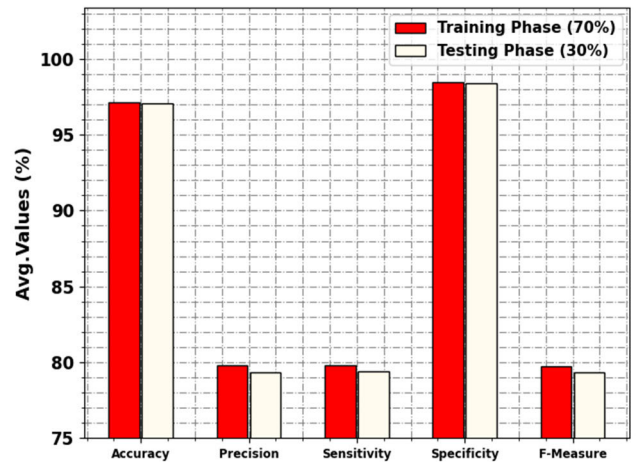
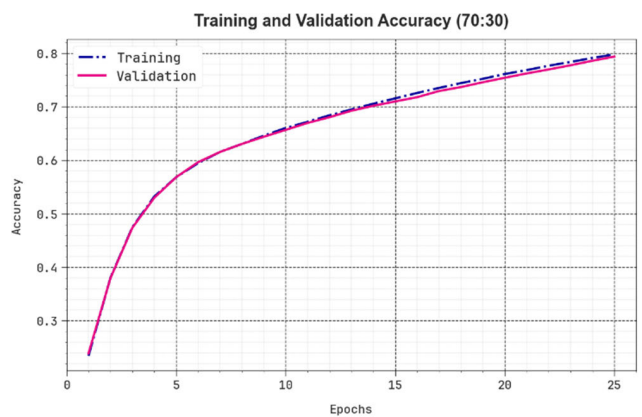
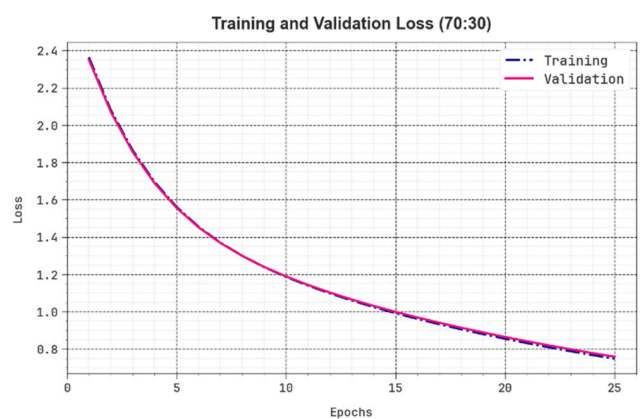
TABLE 4. Attack recognition result of EGTO-FLADC model under 70:30 of TRPH/TSPH.

Classes	$Accu_y$	$Prec_n$	$Sens_y$	$Spec_y$	$F_{Measure}$
TRPH (70%)					
C1	97.09	79.42	80.45	98.38	79.93
C2	97.40	78.81	87.24	98.19	82.81
C3	97.73	82.28	86.33	98.59	84.26
C4	96.69	78.29	73.84	98.44	76.00
C5	97.22	80.69	80.32	98.52	80.50
C6	96.94	78.78	77.82	98.40	78.30
C7	96.87	77.71	78.16	98.30	77.94
C8	96.92	80.52	75.26	98.59	77.80
C9	97.15	79.49	80.19	98.43	79.84
C10	97.12	80.54	79.03	98.52	79.78
C11	96.83	77.87	79.63	98.20	78.74
C12	97.14	81.34	79.21	98.56	80.26
C13	96.93	79.21	75.62	98.52	77.37
C14	97.56	82.16	83.95	98.60	83.05
Average	97.11	79.79	79.79	98.45	79.76
TSPH (30%)					
C1	97.34	80.98	81.09	98.57	81.03
C2	97.34	78.39	86.29	98.19	82.15
C3	97.44	80.43	86.16	98.33	83.20
C4	96.46	77.05	72.80	98.31	74.86
C5	97.17	80.82	78.98	98.56	79.89
C6	96.92	79.60	77.62	98.44	78.60
C7	96.78	77.73	78.54	98.22	78.14
C8	96.76	78.39	74.80	98.43	76.55
C9	97.21	81.52	80.36	98.55	80.94
C10	96.93	77.72	79.19	98.28	78.45
C11	96.99	76.36	78.79	98.28	77.56
C12	97.18	78.56	79.46	98.45	79.01
C13	96.78	80.53	76.31	98.47	78.36
C14	97.43	82.90	80.93	98.71	81.90
Average	97.05	79.36	79.38	98.41	79.33

nature of the EGTO-FLADC method in the pattern detection procedure on both TRA and TES data. The increasing trend in VAL outlines the capability of the EGTO-FLADC technique to adapt to the TRA data and excel in delivering the exact identification of hidden data, indicating strong generalization abilities.

Fig. 11 determines a whole representation of the TRALOS and VALLOS outcomes of the EGTO-FLADC method over distinct epochs under 70:30 of TRPH/TSPH. The advanced reduction in TRALOS highlights the EGTO-FLADC method, enhancing the weights and diminishing the classification error on the TRA and TES data. The figure indicates a clear understanding of the EGTO-FLADC model's association with the TRA data, emphasizing its ability to take patterns within both databases. Remarkably, the EGTO-FLADC model continually improves its parameters to create the differences between the forecast and actual TRA class labels.

The results of inspecting the PR curve, as shown in Fig. 12, show that the EGTO-FLADC technique gradually

**FIGURE 9.** The average outcome of the EGTO-FLADC technique under 70:30 of TRPH/TSPH.**FIGURE 10.** $Accu_y$ curve of EGTO-FLADC technique under 70:30 of TRPH/TSPH.**FIGURE 11.** Loss curve of EGTO-FLADC technique under 70:30 of TRPH/TSPH.

realizes enhanced PR values at each class under 70:30 of TRPH/TSPH. It verifies the boosted skills of the EGTO-FLADC model in classifying separate classes, exhibiting expertise in detecting class labels.

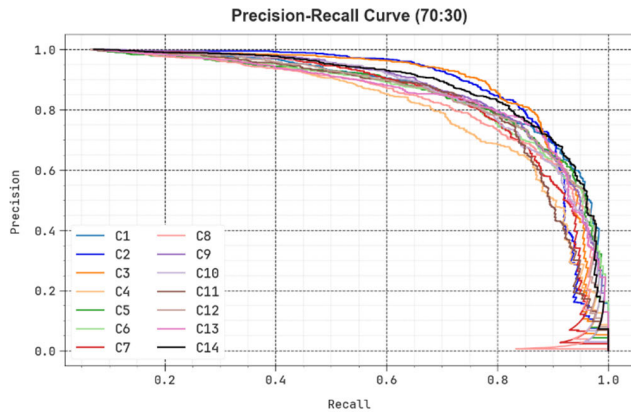


FIGURE 12. PR curve of EGTO-FLADC model under 70:30 of TRPH/TSPH.

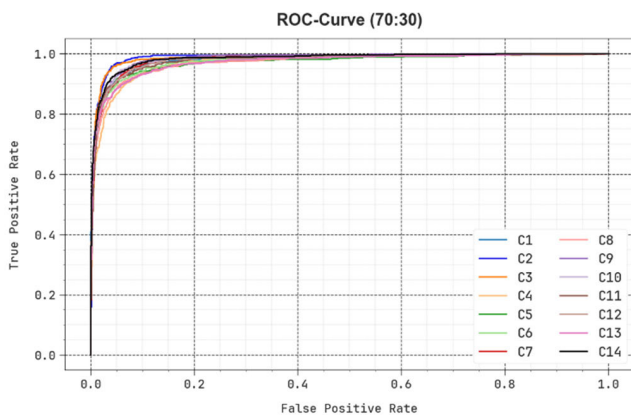


FIGURE 13. ROC curve of EGTO-FLADC model under 70:30 of TRPH/TSPH.

Similarly, in Fig. 13, ROC curves made by the EGTO-FLADC technique outperformed the identification of separate labels under 70:30 of TRPH/TSPH. It explains the trade-off between TPR and FRP over discrete detection threshold values and epoch counts. The outcome highlighted the higher classifier outcomes of the EGTO-FLADC methodology below all classes, outlining the efficacy in tackling several classification issues.

The experimental values of the EGTO-FLADC method are equated with the present techniques in Table 5 and Fig. 14 [19-21-22, 35-37]. The outcomes highlighted the ineffectual recognition results of the Random Forest (RF), Support Vector Machine (SVM), and K-nearest neighbour (KNN) models with the least performance. In line with this, the Adaptive Boosting (AdaBoost) model has shown certainly boosted results. Moreover, the Adaptive Neuro-Fuzzy Inference System (ANFIS) and Decision Tree (DT) models have obtained reasonable results over previous models. Additionally, the GRU, stacked deep polynomial network (S-DPN), Density-Based Spatial Clustering of Applications with Noise (DBSCAN), Kernel-based Online Anomaly Detection (KOAD), and One-class SVM (OCSVM) portrayed a slightly improved value. Nevertheless, the EGTO-FLADC technique

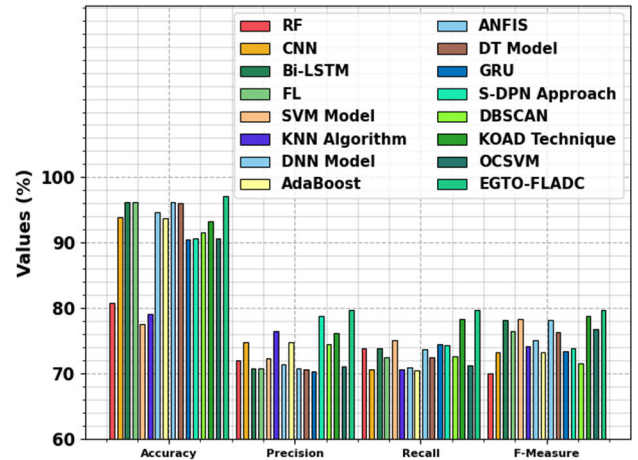


FIGURE 14. Comparative study of EGTO-FLADC model with recent approaches.

has enhanced results with $accu_y$ of 97.11%, $prec_n$ of 79.79%, $reca_l$ of 79.79%, and $F_{measure}$ of 79.76%.

TABLE 5. Comparative study of EGTO-FLADC technique with recent approaches [19-21-22, 35-37].

Models	$Accu_y$	$Prec_n$	$Reca_l$	$F_{Measure}$
RF	80.83	71.99	73.97	69.98
CNN	93.88	74.82	70.59	73.30
Bi-LSTM	96.27	70.84	73.87	78.23
FL	96.16	70.76	72.59	76.48
SVM Model	77.61	72.41	75.16	78.36
KNN				
Algorithm	79.18	76.46	70.69	74.25
DNN Model	94.67	71.49	71.04	75.06
AdaBoost	93.80	74.77	70.54	73.24
ANFIS	96.20	70.79	73.81	78.15
DT Model	96.10	70.69	72.53	76.42
GRU	90.47	70.30	74.54	73.50
S-DPN				
Approach	90.67	78.77	74.34	73.88
DBSCAN	91.56	74.54	72.60	71.66
KOAD				
Technique	93.32	76.20	78.41	78.78
OCSVM	90.63	71.10	71.22	76.89
EGTO-FLADC	97.11	79.79	79.79	79.76

The computational time (CT) analysis in seconds for various techniques used in a specific study is demonstrated in Table 6 and Fig. 15. The RF model takes the longest time at 8.90 seconds, followed closely by the SVM model at 9.33 seconds. KNN and DNN models have comparable computational times of 9.46 seconds each. AdaBoost, known for its efficiency, takes only 3.37 seconds, while the ANFIS model is the fastest at 2.60 seconds. Other techniques like the DT model, GRU, and S-DPN method have times ranging from 4.29 to 7.17 seconds. The DBSCAN method requires 5.30 seconds, while KOAD and OCSVM are slightly slower,

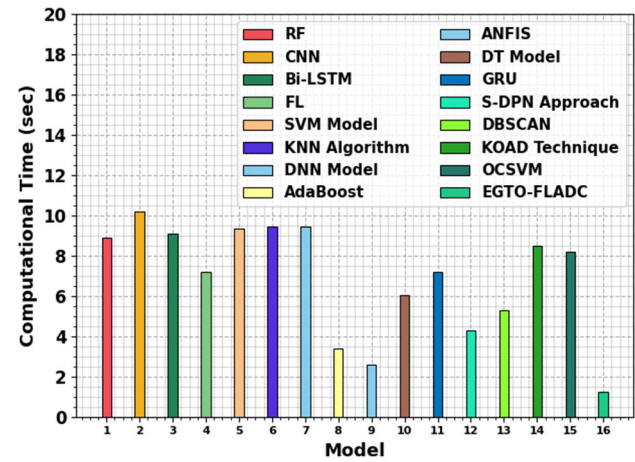


FIGURE 15. CT analysis of EGTO-FLADC model with recent approaches.

taking 8.47 and 8.19 seconds, respectively. The EGTO-FLADC technique is the fastest, with a CT of 1.24 seconds.

TABLE 6. CT analysis of EGTO-FLADC technique with recent approaches.

Model	CT (sec)
RF	8.90
CNN	10.19
Bi-LSTM	9.11
FL	7.19
SVM Model	9.33
KNN Algorithm	9.46
DNN Model	9.46
AdaBoost	3.37
ANFIS	2.60
DT Model	6.06
GRU	7.17
S-DPN Approach	4.29
DBSCAN	5.30
KOAD Technique	8.47
OCSVM	8.19
EGTO-FLADC	1.24

These results show that the EGTO-FLADC technique effectively identifies the attacks in the CIIoT environment.

V. CONCLUSION AND FUTURE WORK

In this study, an EGTO-FLADC approach in a CIIoT environment was proposed. The EGTO-FLADC technique aimed to enhance security in the CIIoT environment via FL and the attack detection process. It involved four major procedures: FL-based ID, min-max-based normalization, TCN-GRU-based attack recognition, and EGTO-based hyperparameter tuning. Primarily, the EGTO-FLADC technique utilized the FL concept for collaborative training of combined models with many devices with no data-sharing process. In addition, the EGTO-FLADC technique utilized a min-max scalar for data normalization. The EGTO-FLADC method employed

the TCN-GRU and the EGTO model for attack recognition for hyperparameter selection. The EGTO model involved the integration of the conventional GTO model with MPA. The performance evaluation of the EGTO-FLADC approach is examined under the EdgeIIoTset dataset. The experimental validation of the EGTO-FLADC approach portrayed a superior accuracy value of 97.11% over existing models.

A. LIMITATIONS AND FUTURE DIRECTIONS

The EGTO-FLADC method has various limitations that need to be addressed for real-world deployment in IIoT security. First, the method assumes ideal conditions for FL, such as reliable communication and uniform data distribution. However, real-world IIoT environments often face challenges like heterogeneous, noisy data and unstable network conditions, which could adversely affect the model’s performance. Furthermore, while the comparative analysis focuses on specific DL models, it overlooks other advanced DL approaches designed explicitly for IIoT security, potentially missing out on more effective baseline comparisons.

Another critical limitation is the scalability of the EGTO-FLADC method in large-scale IIoT networks, where thousands or millions of devices are involved. FL may encounter communication and computational bottlenecks that affect the system’s performance. Moreover, while innovative, the EGTO optimization process introduces extra computational overhead, and a more thorough comparison with simpler optimization methods is needed to evaluate the trade-offs between performance and complexity. The current experiments also do not comprise robustness testing against adversarial attacks or dynamic attack patterns, such as zero-day attacks, common in real-world CIIoT systems.

To address these limitations, future research should focus on improving the scalability and robustness of the EGTO-FLADC method in large-scale and dynamic IIoT networks. Examining the method’s performance in real-world attack scenarios, particularly adversarial manipulation, is significant. Moreover, exploring ways to optimize the integration of FL with BC can improve privacy-preserving capabilities and address challenges related to network latency and decentralized data management. Future studies should also focus on optimizing communication protocols for more efficient decentralized learning, ensuring the EGTO-FLADC method can scale and adapt to the evolving security needs of IIoT systems.

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DATA AVAILABILITY STATEMENT

The data supporting this study's findings are openly available in the Kaggle repository at <https://www.kaggle.com/datasets/mohamedamineferrag/edgeiiotset-cyber-security-dataset-of-iiot-iiot>, reference number [34].

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